# Courier Management System (CMS)

**Project submitted to Thiruvalluvar University, Serkkadu, Vellore.**

**In Partial Fulfilment for the award of degree**

**BACHELOR OF COMPUTER**

**APPLICATION**

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**UG DEPARTMENT OF COMPUTER APPLICATION**

**SHANMUGA INDUSTRIES ARTS AND SCIENCE COLLEGE.**

## User Story:

A courier management organization needs a centralized system to manage its complete delivery process efficiently. The system should allow staff to register and manage shipments, maintain customer and delivery-agent information, assign deliveries to suitable agents, and update shipment statuses throughout the delivery process. Customers should be able to receive accurate information about their parcels and stay informed about delivery progress. Branch managers should be able to monitor shipments, manage branch operations, and view important delivery information from a single platform. The system should also help reduce manual work, improve communication, minimize delivery delays, and provide useful information for monitoring overall courier operations.

# 1. INTRODUCTION

### 1.1 Project Overview :

Courier and logistics companies often face difficulties in managing shipments and deliveries efficiently because of manual operations and disconnected systems. These challenges can result in delayed shipment updates, limited visibility for customers, and reduced operational efficiency.

The **Courier Management System (CMS)** is a Salesforce-based cloud solution designed to centralize and simplify courier operations. The system supports shipment booking, shipment tracking, delivery agent assignment, automated status updates, billing, and analytics through a unified platform. By integrating these activities into a single system, CMS helps improve operational efficiency, enhance customer satisfaction, and provide real-time business insights for effective management and decision-making.

### 1.2 Purpose :

The main purpose of the Courier Management System is to provide a structured platform for handling courier activities from shipment creation to final delivery. The system is intended to make it easier for employees to maintain customer details, shipment records, delivery information, and payment-related data in an organized manner. It also helps staff monitor the progress of parcels and maintain updated delivery information. By bringing important courier activities into one Salesforce environment, the system reduces dependence on paperwork and separate records. The platform also provides managers with organized information that can be used to monitor daily operations and improve the management of courier services.

## 2. IDEATION PHASE

### 2.1 Problem Statement (Customer Problem Statement Template)

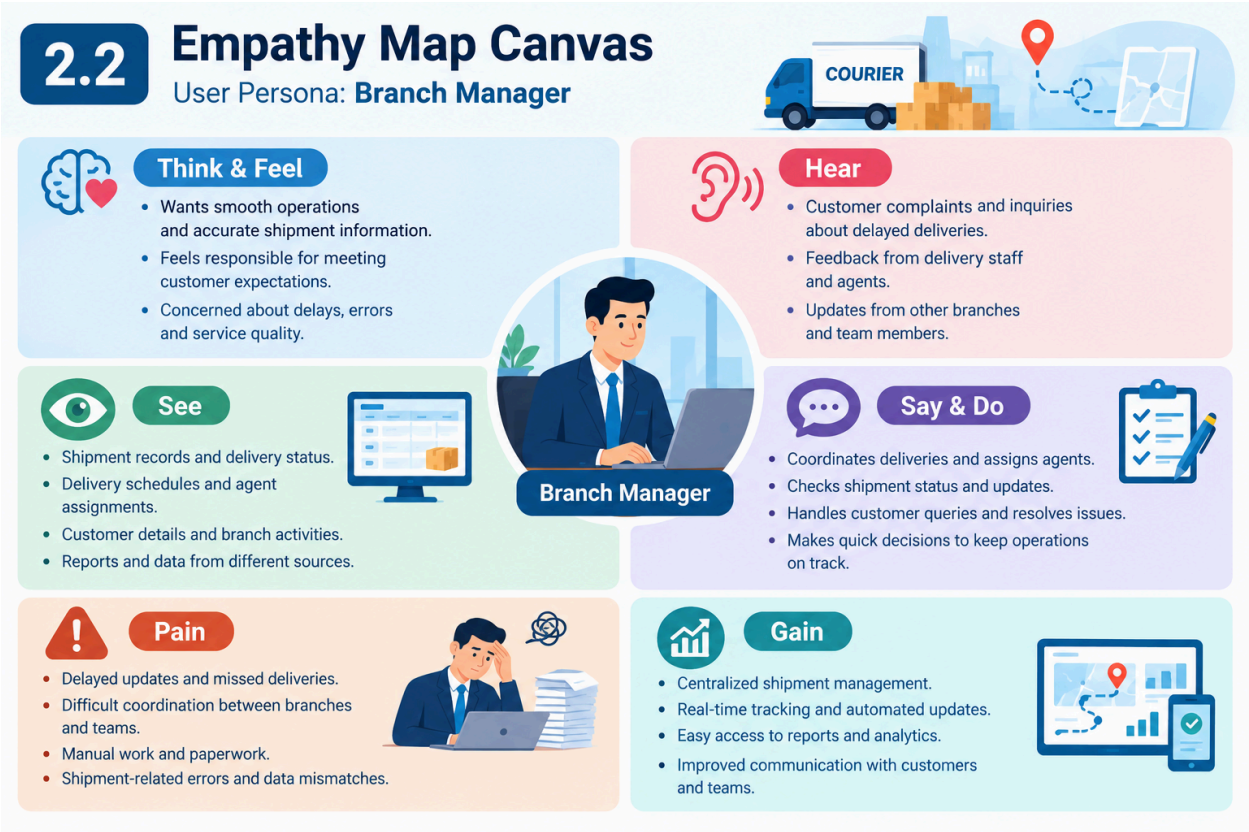
| **PS** | **I am (Customer)** | **I'm trying to** | **But** | **Because** | **Which makes me feel** |
|---|---|---|---|---|---|
| PS-1 | Branch Manager | manage shipments efficiently | tracking is manual | systems are disconnected | stressed |
| PS-2 | Customer | track parcels | updates are delayed | no centralized system | uncertain |




## 2.2 Empathy Map Canvas

**User Persona:** Branch Manage

- **Think & Feel:** Wants smooth operations and accurate shipment information.
- **Hear:** Requests and feedback from customers and delivery staff.
- **See:** Shipment records, delivery schedules, customer details, and branch activities.
- **Say & Do:** Coordinates deliveries, assigns agents, checks shipment status, and handles customer queries.
- **Pain:** Delayed updates, difficult coordination, manual work, and shipment-related errors.
- **Gain:** Centralized shipment management, easy tracking, automated updates, and useful reports.




## 2.3 Brainstorming & Idea Prioritization

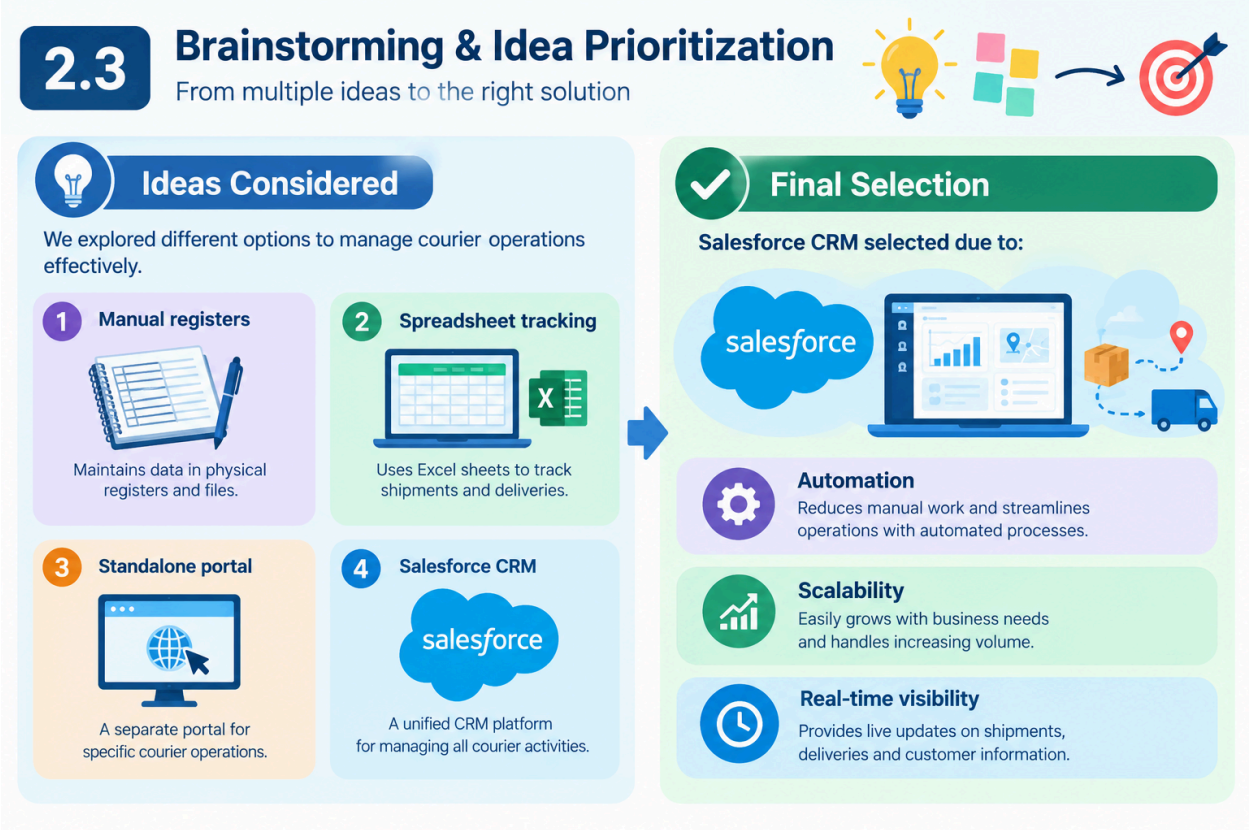
### Ideas Considered

- Manual registers
- Spreadsheet tracking
- Standalone portal
- Salesforce CRM

### Final Selection

Salesforce CRM selected due to:

- Automation
- Scalability
- Real-time visibility

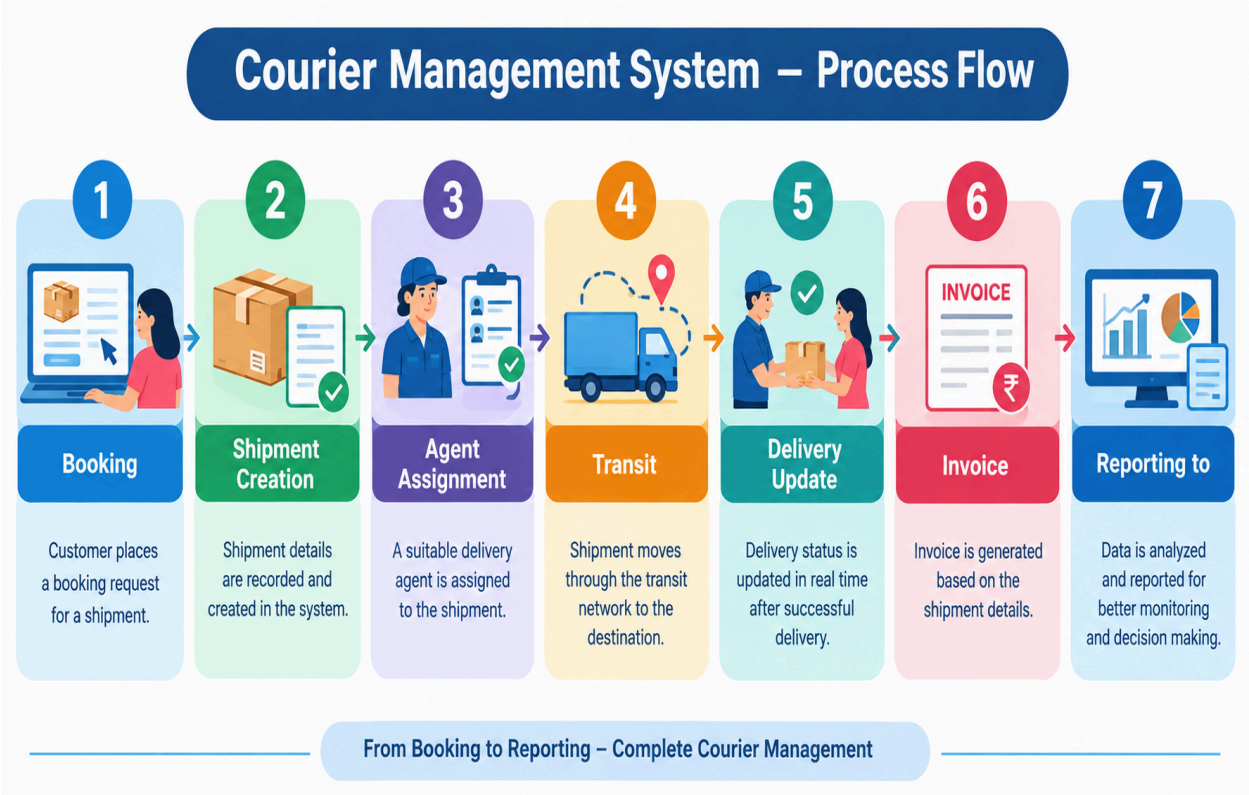



# 3.REQUIREMENT ANALYSIS PHASE

## 3.1 Customer Journey Map

Stages:

- Booking → Shipment Creation → Agent Assignment → Transit → Delivery Update → Invoice → Reporting




## 3.2 Solution Requirements

### Functional Requirements

| FR No | Requirement |
|---|---|
| FR-1 | Customer management |
| FR-2 | Shipment booking |
| FR-3 | Delivery agent assignment |
| FR-4 | Real-time status updates |
| FR-5 | Invoice generation |
| FR-6 | Reports & dashboards |

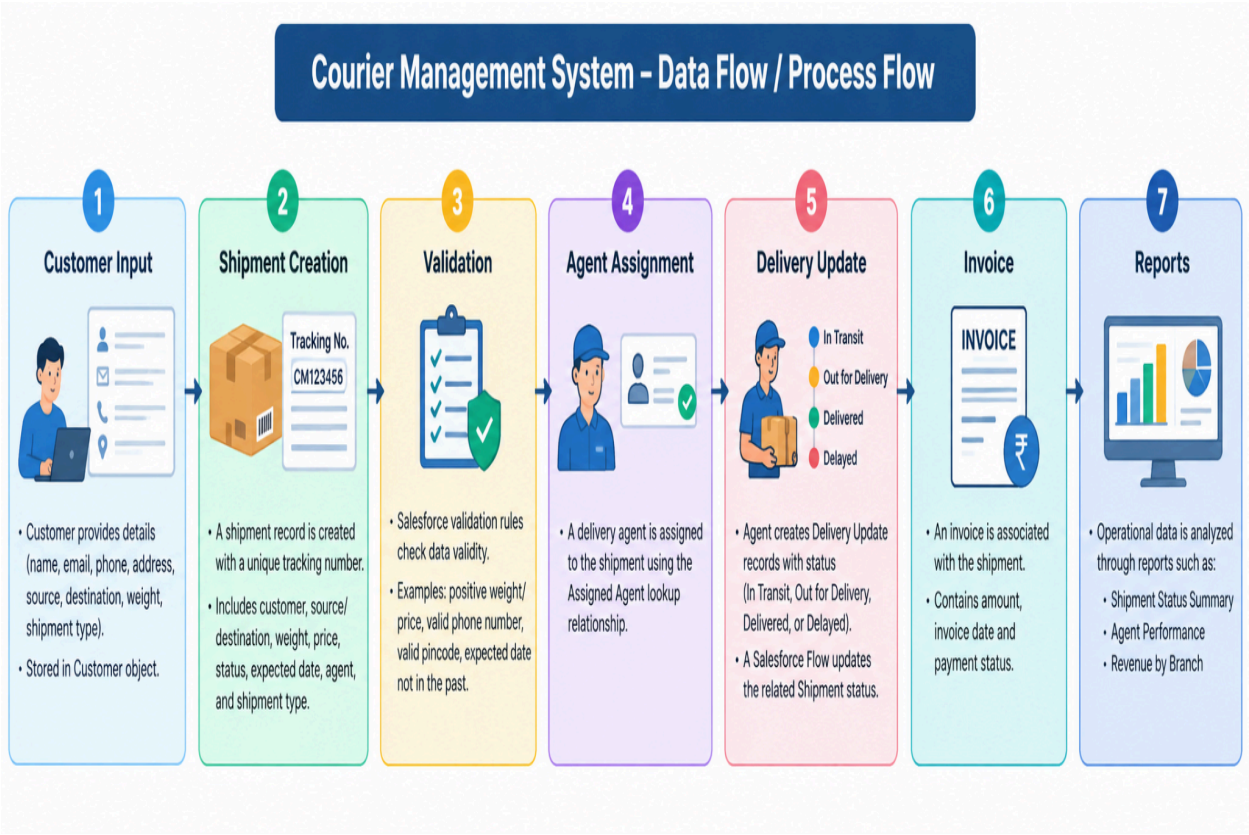
**Non-Functional Requirements**

| NFR | Description |
|---|---|
| Usability | Easy Lightning interface |
| Security | Role-based access control |
| Reliability | Error-free automation |
| Performance | Fast tracking updates |
| Availability | 24/7 cloud access |
| Scalability | Supports growing branches |

## 3.3 Data Flow Diagram

Flow:

Customer Input → Shipment Creation → Validation → Agent Assignment → Delivery Update → Invoice → Reports



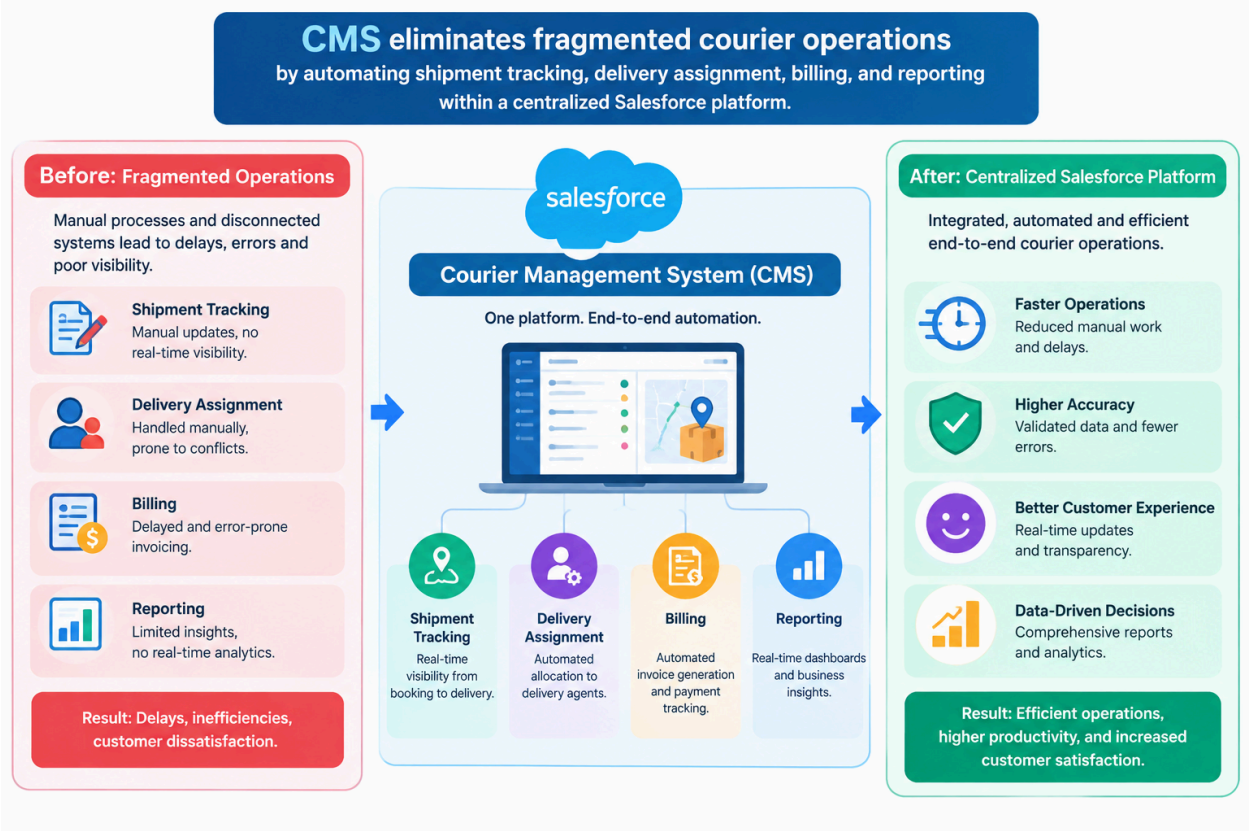

### 3.4 Technology Stack

| **Layer** | **Technology** |
|---|---|
| UI | Salesforce Lightning |
| Logic | Salesforce Flows |
| Database | Salesforce Objects |
| Automation | Validation Rules, Flows |
| Security | Profiles, Roles |
| Reporting | Reports & Dashboards |

# 4. PROJECT DESIGN PHASE

## 4.1 Problem–Solution Fit

CMS eliminates fragmented courier operations by automating shipment tracking, delivery assignment, billing, and reporting within a centralized Salesforce platform.



## 4.2 Proposed Solution

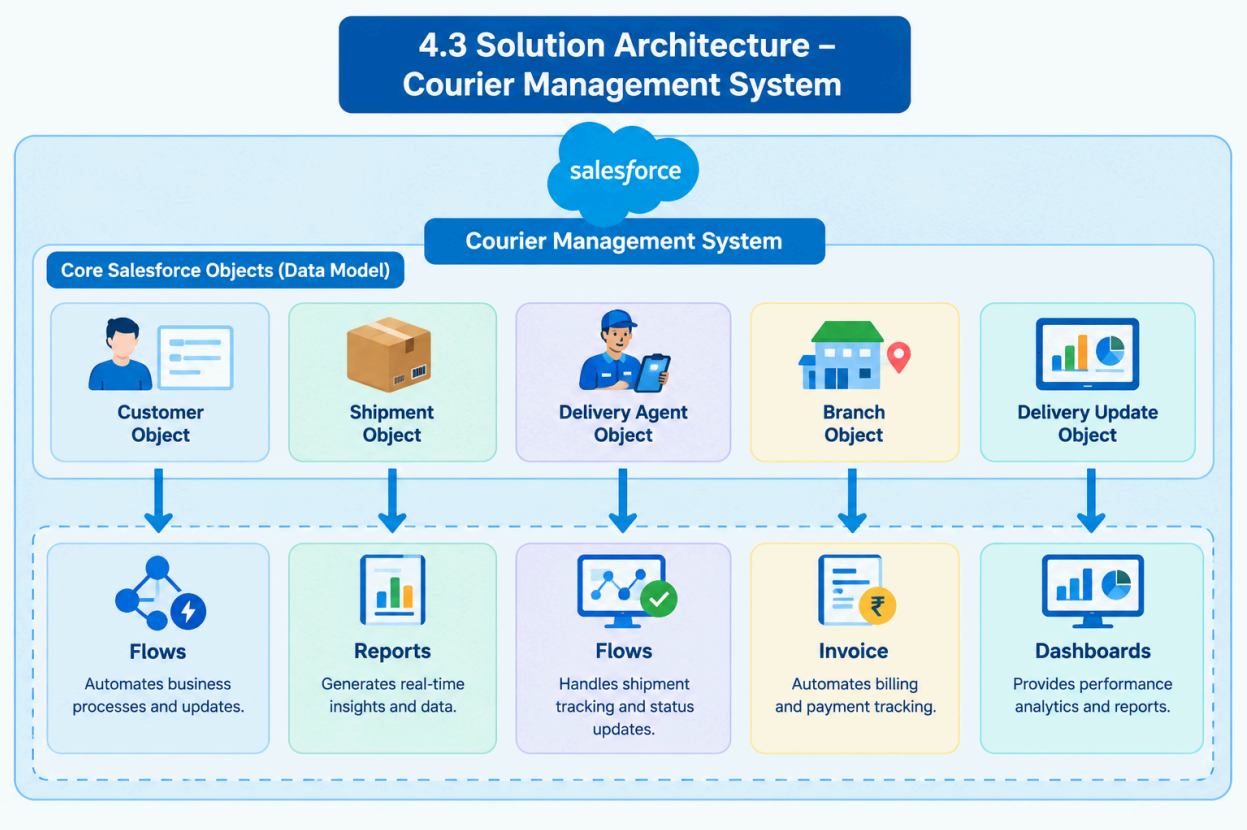
### Functional Requirements

| Parameter | Description |
|---|---|
| Problem | Manual shipment management |
| Solution | Salesforce CMS |
| Innovation | Automated tracking & billing |
| Impact | Faster deliveries |
| Business Model | Courier services |
| Scalability | Multi-branch ready |

## 4.3 Solution Architecture

Components:

- Customer Object
- Shipment Object
- Delivery Agent Object
- Branch Object
- Delivery Update Object
- Invoice Object
- Flows
- Reports
- Dashboards




# 5. PROJECT PLANNING & SCHEDULING

## 5.1 Project Planning

- Agile methodology
- Epics → Stories → Story Points
- Sprint-based execution
- Velocity calculation used for estimation

**Product Backlog, Sprint Schedule, and Estimation (4 Marks)**

Use the below template to create product backlog and sprint schedule

| Sprint | Functional Requirement (Epic) | User Story Number | User Story / Task | Story Points | Priority | Team Members |
|---|---|---|---|---|---|---|
| Sprint-1 | Developer Setup | USN-1 | As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Courier Management System. | 3 | High | Member1 |
| Sprint-2 | Data Modeling | USN-2 | As an admin, I want to create Customer, Shipment, Delivery | 5 | High | Member2 |

| Sprint | Functional Requirement (Epic) | User Story Number | User Story / Task | Story Points | Priority | Team Members |
|---|---|---|---|---|---|---|
| | | | Agent, Branch, and Invoice objects with proper relationships so that courier data is stored and managed efficiently. | | | |
| Sprint-2 | Data Modeling | USN-3 | As a staff member, I want tabs and a Lightning App for all courier modules so that I can easily navigate and perform daily operations. | 5 | High | Member3 |
| Sprint-3 | Automation | USN-4 | As a system, I want to validate shipment details such as delivery dates, addresses, and mandatory fields so that incorrect or incomplete records are prevented. | 3 | High | Member4 |
| Sprint-3 | Automation | USN-5 | As a dispatcher, I want automated flows for shipment status updates and agent assignment so that deliveries are processed faster without manual intervention. | 3 | High | Member4 |
| Sprint-4 | Security | USN-6 | As an administrator, I want role-based access control for managers, agents, and staff so that sensitive shipment and billing data is secure. | 5 | High | Member2 |
| Sprint-5 | Reports | USN-7 | As a branch manager, I want reports showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions. | 4 | High | Member3 |
| Sprint-6 | Dashboards | USN-8 | As a branch manager, I want dashboards showing delivery performance, shipment status, and revenue so that I can monitor operations and make informed decisions. | 4 | Medium | Member4 |

**Project Tracker, Velocity & Burndown Chart: (4 Marks)**

| Sprint | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed (as on Planned End Date) | Sprint Release Date (Actual) |
|---|---|---|---|---|---|---|
| Sprint-1 | 20 | 6 Days | 24 Oct 2022 | 29 Oct 2022 | 20 | 29 Oct 2022 |

| Sprint | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed (as on Planned End Date) | Sprint Release Date (Actual) |
|---|---|---|---|---|---|---|
| Sprint-2 | 20 | 6 Days | 31 Oct 2022 | 05 Nov 2022 | | |
| Sprint-3 | 20 | 6 Days | 07 Nov 2022 | 12 Nov 2022 | | |
| Sprint-4 | 20 | 6 Days | 14 Nov 2022 | 19 Nov 2022 | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |
| | | | | | | |

### 6. Project Development Phase:

**What you'll learn**

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Developing reports & dashboards
7. Setting up security with profiles, permission sets, and sharing
8. End-to-end application testing

# Milestone 1: Salesforce Account

### Introduction:

New to Salesforce and unsure where to start? Curious how real business applications—like loan processing and payment tracking—can be built on the platform? You’re in the right place.

Welcome to the Loan Management System project!

Salesforce offers powerful tools that help automate processes, improve accuracy, and manage customer operations with ease. In this project, you'll learn how Salesforce can streamline lending workflows—from borrower onboarding to loan approval, EMI tracking, and document verification.

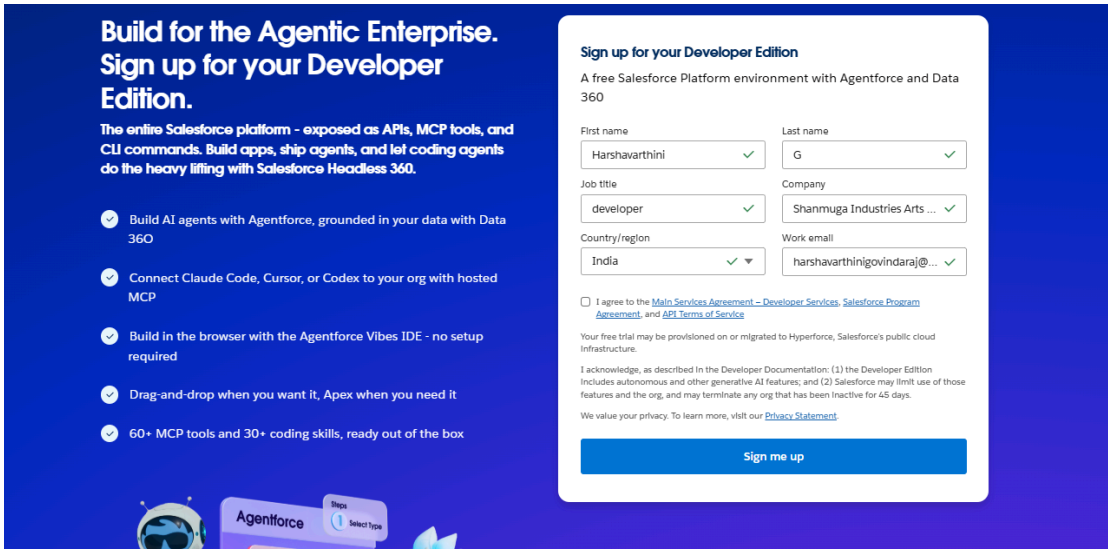
## What Is Salesforce ?

Salesforce is your customer success platform, designed to help you sell, service, market, analyze, and connect with your customers. Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the cloud.

### Activity 1: Creating Developer Account

Creating a developer org in salesforce.

1. Go to https://developer.salesforce.com/signup
2. On the sign up form, enter the following details :
3. First name & Last name
4. Email




1) First name & Last name

2) Email

3) Job Title: Developer

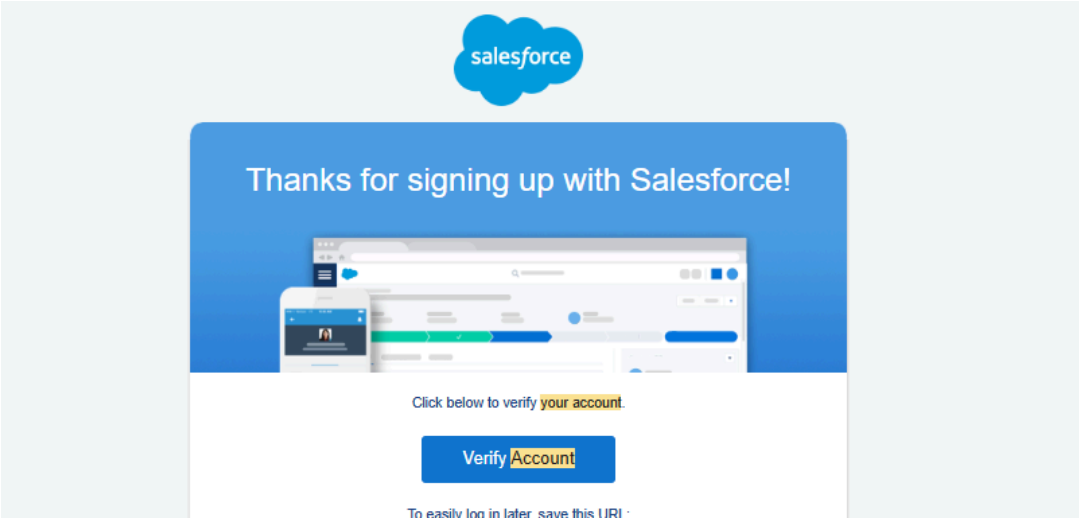
4) Company: College Name

5) Country: India

Click on sign me up after filling these.

**Activity 2: Account Activation**

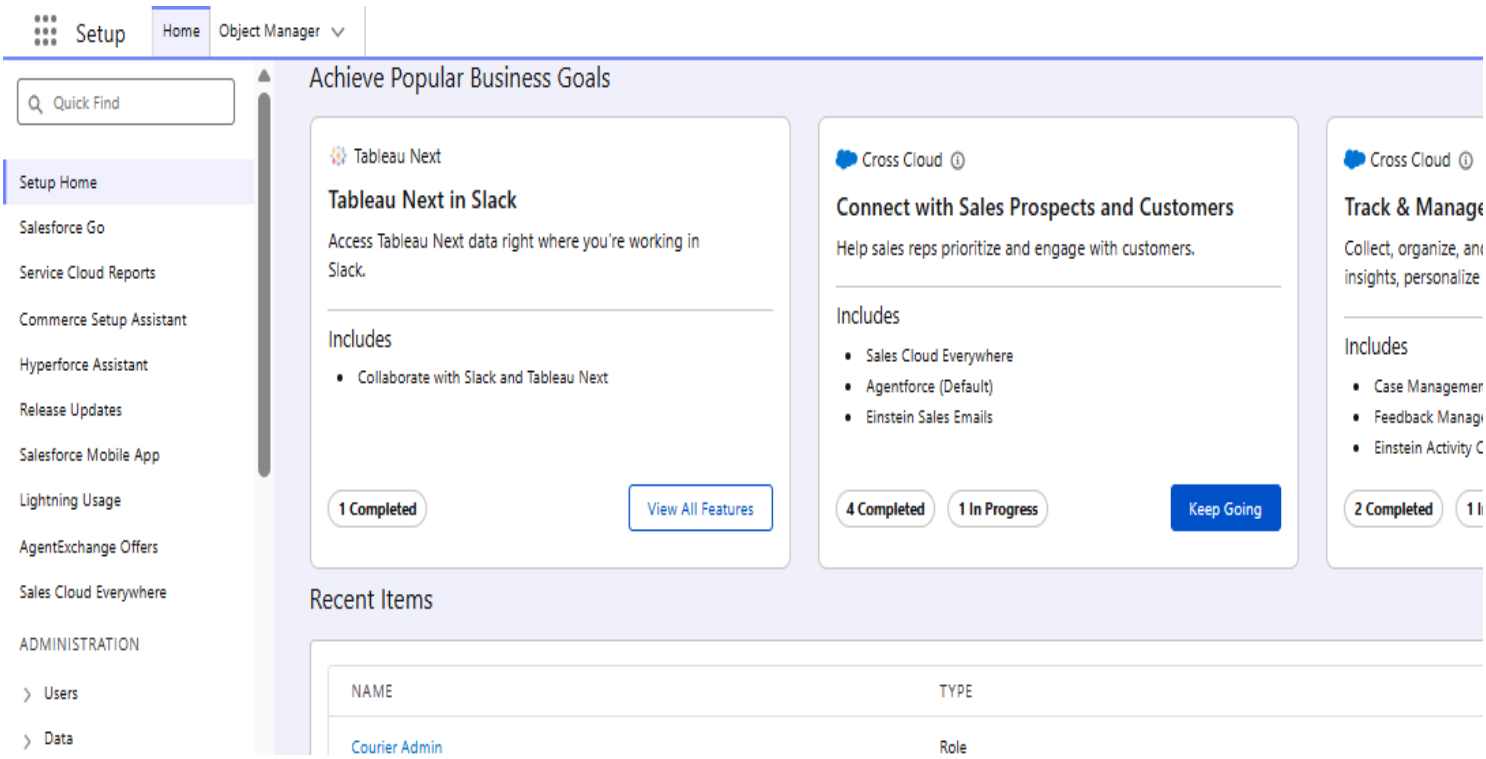
1. Go to the inbox of the email that you used while signing up. Click on the verify account to activate your account. The email may take 10-30 mins and sometimes 2 hours.




2. Click on Verify Account
3.Give a password and answer a security question and click on change password.
4.Then you will redirect to your salesforce setup page.

# Milestone 2- Object:

## What is an object?




Salesforce objects are database tables that permit you to store data that is specific to an organization. It consists of fields (columns) and records (rows).

### Custom Objects Created:

1. **Customer__c** – stores customer personal and contact information
2. **Shipment__c** – manages parcel details, source, destination, and tracking information
3. **Delivery_Agent__c** – maintains delivery agent profiles and branch details
4. **Branch__c** – represents courier branch offices
5. **Delivery_Update__c** – stores status updates for each shipment
6. **Invoice__c** – manages billing details for delivered shipments

**Relationships:**

Customer → Shipment (Lookup)

Shipment → Delivery_Agent (Lookup)

Delivery_Agent → Branch (Lookup)

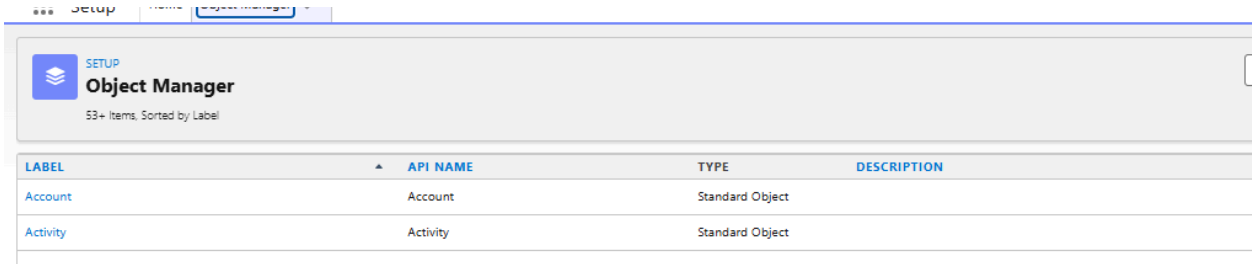
Delivery_Update → Shipment (Master-Detail)

Invoice → Shipment (Lookup)

# Steps to Implement :

## Activity 1: Create Customer Object

1. Click Setup → Object Manager → Create → Custom Object




2. Enter the following details:

- Label: Customer
- Plural Label: Customers
- Record Name: Customer Name (Text)

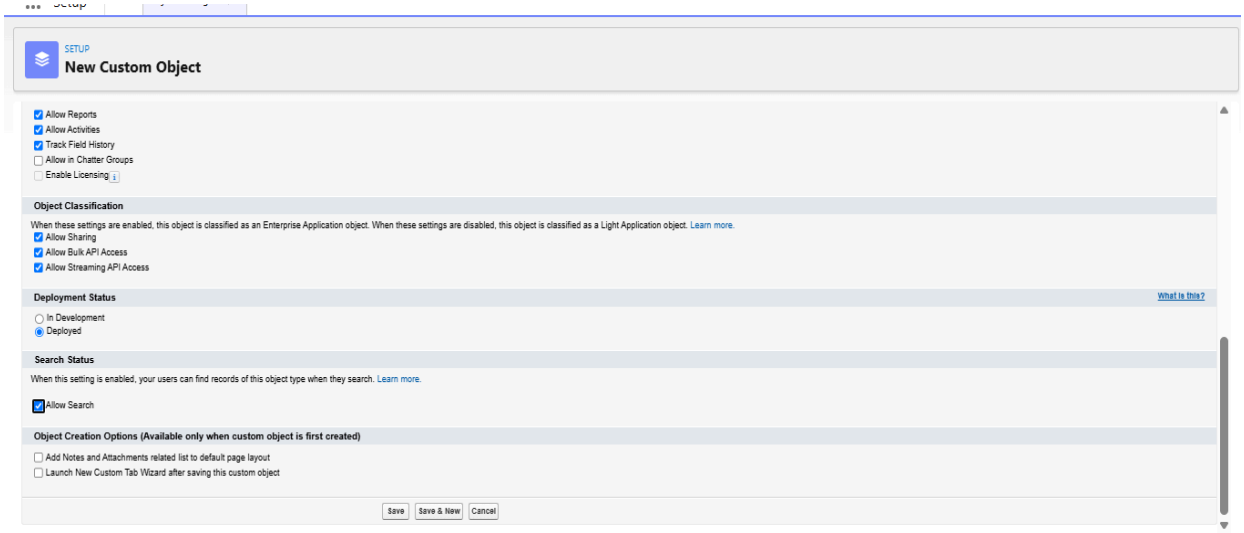
3. Under Optional Features:

- Enable Allow Reports
- Enable Track Field History

- In Search Status check the box Allow Search

- In Object Creation Options check the box Add Notes and Attachments related list to default page layout

4. Click Save.




## Activity 2: Create Shipment Object

1. Go to Setup → Object Manager → Create → Custom Object

2. Enter:

- Label: Shipment

- Plural Label: Shipments

- Record Name: Tracking Number (Auto Number)

3. Example Format: SHP-{00000}

   Starting Number: 0

4. Enable:

- Allow Reports

- Track Field History

5. Click Save

### Activity 3: Create Delivery_Agent Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:

- Label: Delivery Agent

- Plural Label: Delivery Agents

- Record Name: Agent Name (Text)

3. Enable:

- Allow Reports

- Track Field History

4. Click Save

### Activity 4: Create Branch Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:

- Label: Branch

- Plural Label: Branches

- Record Name: Branch Name (Text)

3. Enable:
- Allow Reports

4. Click Save.

## Activity 5: Create Delivery_Update Object

1. Setup → Object Manager → Create → Custom Object

2. Enter:
- Label: Delivery Update

- Plural Label: Delivery Updates

- Record Name: Update Number (Auto Number)

3. Example Format: UPD-{0000}
   Enter Starting number : 0
4. Enable:
- Allow Reports

- Track Field History

5. Click Save.

### Activity 6: Create Invoice Object

1. Setup → Object Manager → Create → Custom Object
2. Enter:

- Label: Invoice
- Plural Label: Invoices
- Record Name: Invoice Number (Auto Number)

3. Example Format: INV-{0000}
   Enter Starting number: 0
4. Enable:

- Allow Reports
- Track Field History

5. Click Save

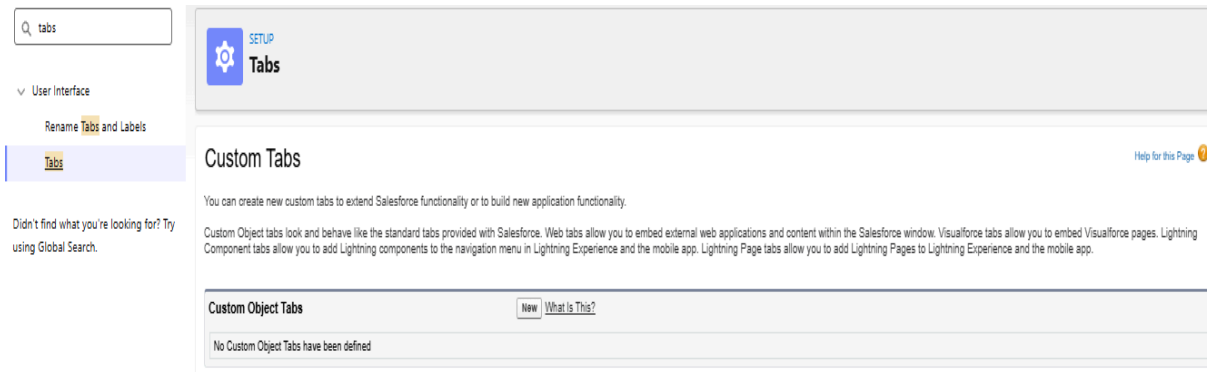
## Milestone 3 - Tabs

**Tab:**

Tabs in Salesforce help users view the information at a glance. It displays the data of objects and other web content in the application.

### Activity 1: Creating a Tab for Customer Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.




3. For Object, select Customer.
4. For Tab Style, choose any icon.
5. Leave all default settings as they are.
6. Click Next, Next, and Save.

## Activity 2: Creating a Tab for Shipment Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Shipment.
4. Choose any Tab Style.
5. Keep defaults unchanged.
6. Click Next, Next, and Save.

## Activity 3: Creating a Tab for Delivery Agent Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.

3. For Object, select Delivery Agent.
4. Pick any Tab Style.
5. Leave default settings.
6. Click Next, Next, and Save.

## Activity 4: Creating a Tab for Branch Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Branch.
4. Choose any Tab Style.
5. Keep all defaults.
6. Click Next, Next, and Save.

## Activity 5: Creating a Tab for Delivery Update Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Delivery Update.
4. Select any Tab Style.
5. Keep default configurations.
6. Click Next, Next, and Save.

# Activity 6: Creating a Tab for Invoice Object

1. Click Home, type Tabs in Quick Find, and select Tabs.
2. Under Custom Object Tabs, click New.
3. For Object, select Invoice.
4. Select any Tab Style.
5. Leave defaults unchanged.
6. Click Next, Next, and Save.

## Milestone 4: Fields & Relationships

**What are fields ?**

Fields in Salesforce represent what the columns represent in relational databases. It can store data values which are required for a particular object in a record.

# Activity 1: Creation of fields for the Customer Object

| Field Name | API Name | Data Type | Required | Description |
|---|---|---|---|---|
| Customer Name | Name | Text | Yes | Full name of customer |
| Email | Email__c | Email | Yes | Customer email |
| Phone | Phone__c | Phone | Yes | Customer contact number |
| Address | Address__c | Text Area | Yes | Customer address |

| Customer Type | Customer_Type__c | Picklist | Yes | Regular / Business |
|---|---|---|---|---|
| City | City__c | Text | No | Customer city |
| State | State__c | Text | No | Customer state |
| Pincode | Pincode__c | Text | No | Postal code |
| Is Active | Is_Active__c | Checkbox | No | Active status |

### 1. Email (Email Field)

- Go to Setup → Object Manager → Customer__c
- Click Fields & Relationships → New
- Select Email → Next
- Field Label: Email
- Check the Required box
- Click Next → Next → Save

### 2. Phone (Phone Field)

- Setup → Object Manager → Customer__c
- Fields & Relationships → New
- Select Phone → Next
- Field Label: Phone
- Check Required
- Click Next → Next → Save

### 3. Address (Text Area)

- Setup → Object Manager → Customer__c

- Fields & Relationships → New
- Select Textarea → Next
- Field Label: Address
- Check Required
- Save

### 4. Customer Type (Picklist)

- Setup → Customer__c → Fields → New
- Picklist → Next
- Field Label: Customer Type
- Enter values:
    - Regular
    - Business
- Check Required
- Save

### 5. City (Text)

- Setup → Customer__c → Fields → New
- Text → Next
- Field Label: **City**
- Save

### 6. State (Text)

- Setup → Customer__c → Fields → New
- Text → Next
- Field Label: **State**
- Save

### 7. Pincode (Text)

- Setup → Customer__c → Fields → New
- Text → Next
- Field Label: **Pincode**
- Save

### 8. Is Active (Checkbox)

- Setup → Customer__c → Fields → New
- Checkbox → Next
- Field Label: **Is Active**
- Save

# Activity 2: Creation of fields for the Shipment Object

| Field Name | API Name | Data Type | Required | Description |
|---|---|---|---|---|
| Tracking Number | Name | Auto Number | Yes | Unique ID |
| Customer | Customer__c | Lookup(Customer) | Yes | Related customer |
| Source Address | Source_Address__c | Text Area | Yes | Pickup location |
| Destination Address | Destination_Address__c | Text Area | Yes | Delivery location |
| Weight | Weight__c | Number | Yes | Parcel weight |
| Price | Price__c | Currency | Yes | Shipping charge |
| Status | Status__c | Picklist | Yes | Shipment status |
| Expected Date | Expected_Date__c | Date | Yes | Delivery estimate |
| Assigned Agent | Assigned_Agent__c | Lookup(Delivery_Agent) | No | Delivery agent |
| Shipment Type | Shipment_Type__c | Picklist | Yes | Domestic/International |

### 1. Source Address (Text Area)

- Setup → Shipment__c → Fields → New
- Textarea → Next
- Field Label: **Source Address**

- Required → Save

### 2. Destination Address (Text Area)

- Setup → Shipment__c → Fields → New
- Textarea → Next
- Field Label: **Destination Address**
- Required → Save

### 3. Weight (Number)

- Setup → Shipment__c → Fields → New
- Number → Next
- Field Label: **Weight**
- Required → Save

### 4. Price (Currency)

- Setup → Shipment__c → Fields → New
- Currency → Next
- Field Label: **Price**
- Required → Save

### 5. Status (Picklist)

Values:

- Booked
- In Transit
- Out for Delivery
- Delivered
- Cancelled

Required → Save

### 6. Expected Date (Date)

- Setup → Shipment__c → Fields → New
- Date → Next
- Field Label: **Expected Date**
- Required → Save

### 7. Shipment Type (Picklist)

Values:

- Domestic
- International

Required → Save

# Activity 3: Creation of fields for Delivery_Agent Object

| Field Name | API Name | Data Type | Required |
|---|---|---|---|
| Agent Name | Name | Text | Yes |
| Email | Email__c | Email | Yes |
| Phone | Phone__c | Phone | Yes |
| Branch | Branch__c | Lookup | Yes |
| Status | Status__c | Picklist | Yes |
| Joining Date | Joining_Date__c | Date | No |

### 1. Email

- Setup → Delivery_Agent__c → Fields → New
- Email → Next
- Field Label: Email
- Required → Save

### 2. Phone

- Setup → Delivery_Agent__c → Fields → New
- Phone → Next
- Field Label: Phone
- Required → Save

### 3. Status (Picklist)

Values:

- Active
- Inactive
- On Leave

Required → Save

### 4. Joining Date

- Setup → Delivery_Agent__c → Fields → New
- Date → Next
- Field Label: Joining Date
- Save

# Activity 4: Creation of fields for Branch Object

| Field Name | API Name | Data Type | Required |
|---|---|---|---|
| Branch Name | Name | Text | Yes |
| Branch Code | Branch_Code__c | Text | Yes |
| City | City__c | Text | Yes |
| State | State__c | Text | Yes |
| Manager Name | Manager_Name__c | Text | No |

**Create each field using:**

- Setup → Branch__c → Fields → New

- Select respective Data Type
- Enter Field Label
- Save

(Repeat for all fields)

## Activity 5: Creation of fields for Delivery_Update Object

| Field Name | API Name | Data Type | Required |
|---|---|---|---|
| Update Number | Name | Auto Number | Yes |
| Shipment | Shipment__c | Master-Detail | Yes |
| Status | Status__c | Picklist | Yes |
| Update Date | Update_Date__c | DateTime | Yes |
| Remarks | Remarks__c | Text | No |

**Status Picklist Values:**

- In Transit
- Out for Delivery
- Delivered
- Delayed

Required → Save

## Activity 6: Creation of fields for Invoice Object

| Field Name | API Name | Data Type | Required |
|---|---|---|---|
| Invoice Number | Name | Auto Number | Yes |
| Shipment | Shipment__c | Lookup | Yes |
| Amount | Amount__c | Currency | Yes |
| Invoice Date | Invoice_Date__c | Date | Yes |
| Payment Status | Payment_Status__c | Picklist | Yes |

**Payment Status Values**

- Paid
- Unpaid
- Pending

Required → Save

## Relationships :

| Child Object | Field | Parent Object | Type |
|---|---|---|---|
| Shipment | Customer__c | Customer | Lookup |
| Shipment | Assigned_Agent__c | Delivery_Agent | Lookup |
| Delivery_Agent | Branch__c | Branch | Lookup |
| Delivery_Update | Shipment__c | Shipment | Master-Detail |

| | | | |
|---|---|---|---|
| Invoice | Shipment__c | Shipment | Lookup |

# Relationship Creation Steps :

### Activity 1 – Shipment → Customer

- Setup → Shipment__c → Fields → New
- Lookup Relationship
- Related Object: **Customer__c**
- Field Label: **Customer**
- Save

### Activity 2 – Shipment → Delivery Agent

- Setup → Shipment__c
- New → Lookup
- Object: **Delivery_Agent__c**
- Label: **Assigned Agent**
- Save

### Activity 3 – Delivery Agent → Branch

- Setup → Delivery_Agent__c
- New → Lookup
- Related Object: **Branch__c**

- Label: **Branch**
- Save

### Activity 4 – Delivery Update → Shipment

- Setup → Delivery_Update__c
- New → **Master-Detail**
- Related Object: **Shipment__c**
- Save

### Activity 5 – Invoice → Shipment

- Setup → Invoice__c
- New → Lookup
- Object: **Shipment__c**
- Save

## Milestone 5-Validation Rules :

Validation rules are applied when a user tries to save a record and are used to check if the data meets specified criteria. If the criteria are not met, the validation rule triggers an error message and prevents the user from saving the record until the issues are resolved.

# Activity 1: Create Validation Rules on Customer__c

### 1. Validation Rule: Email Must Not Be Blank

**Steps:**

- Click the Gear icon → Setup
- Go to Object Manager
- Search and select Customer__c
- From the left menu, click Validation Rules
- Click New
- Rule Name: Email_Required
- Check Active

In Error Condition Formula, paste:

ISBLANK(Email__c)

- Click Check Syntax
- Error Message: Email is required.
- Error Location: Field → Email
- Click Save.

### 2. Validation Rule: Phone Must Be 10 Digits

**Steps:**

- Setup → Object Manager → Customer__c
- Validation Rules → New

- Rule Name: Validate_Phone_Number
- Check Active

**Formula:**

LEN(Phone__c) <> 10

- Check Syntax
- Error Message: **Phone number must be 10 digits.**
- Error Location: Field → Phone
- Save

### 3. Validation Rule: Pincode Must Be 6 Digits

**Steps:**

- Setup → Object Manager → Customer__c
- Validation Rules → New
- Rule Name: Validate_Pincode
- Active → Yes

**Formula:**

LEN(Pincode__c) <> 6

- Error Message: **Pincode must be 6 digits.**
- Location: Field → Pincode

- Save

# Activity 2: Create Validation Rules on Shipment__c

### 1. Validation Rule: Weight Must Be Greater Than Zero

**Steps:**

- Setup → Object Manager → Shipment__c
- Click Validation Rules → New
- Rule Name: Validate_Weight
- Active → Checked

**Formula:**

Weight__c <= 0

- Check Syntax
- Error Message: **Weight must be greater than zero.**
- Location: Field → Weight
- Save

### 2. Validation Rule: Price Must Be Greater Than Zero

**Steps:**

- Setup → Object Manager → Shipment__c
- Validation Rules → New
- Rule Name: Validate_Price

- Active → Yes

**Formula:**

Price__c <= 0

- Error Message: **Price must be greater than zero.**
- Location: Field → Price
- Save

### 3. Validation Rule: Expected Date Cannot Be in the Past

**Steps:**

- Setup → Object Manager → Shipment__c
- Validation Rules → New
- Rule Name: Validate_Expected_Date

**Formula:**

Expected_Date__c < TODAY()

- Check Syntax
- Error: **Expected delivery date cannot be in the past.**
- Location: Field → Expected Date
- Save

# Activity 3: Create Validation Rules on Delivery_Agent__c

### 1. Validation Rule: Email Must Be Valid

**Steps:**

- Setup → Object Manager → Delivery_Agent__c
- Validation Rules → New
- Rule Name: Validate_Agent_Email
- Active → Yes

**Formula:**

NOT(CONTAINS(Email__c, "@"))

- Error Message: **Enter a valid email address.**
- Location: Field → Email
- Save

# Activity 4: Create Validation Rules on Branch__c

### 1. Validation Rule: Branch Code Must Not Be Blank

**Steps:**

- Setup → Object Manager → Branch__c
- Validation Rules → New
- Rule Name: Branch_Code_Required

**Formula:**

ISBLANK(Branch_Code__c)

- Error Message: **Branch Code is mandatory.**
- Location: Field → Branch Code
- Save

# Activity 5: Create Validation Rules on Delivery_Update__c

### 1. Validation Rule: Remarks Required If Status is Delayed

**Steps:**

- Setup → Object Manager → Delivery_Update__c
- Validation Rules → New
- Rule Name: Remarks_Required_For_Delay

**Formula:**

```
AND(
 ISPICKVAL(Status__c, "Delayed"),
 ISBLANK(Remarks__c)
)
```

- Error Message: **Remarks are required when status is Delayed.**
- Location: Field → Remarks
- Save

# Milestone 6 - Page Layouts :

Page Layouts in Salesforce control what a user sees on a record page and how information is organized.
They help structure fields, sections, buttons, related lists, and other components in a clean and user-friendly way.
By customizing page layouts, we ensure that users can easily enter, update, and view the required data.

## Activity 1: Edit Page Layout for Customer__c

**Steps:**

- Setup → Object Manager → Customer__c
- Click Page Layouts
- Click the default layout (usually Customer Layout)

**Sections to Add & Arrange:**

- Open the default layout
- In the top palette, click “Section”
- Drag the Section and drop it anywhere on the layout

**A pop-up appears → Fill details:**

- Section Name: “Customer Information”
- Layout:
    - 1-Column or
    - 2-Column (recommended)

- Tab-Key Order: Top-Down
- Click OK

The new section will appear on the layout.

- Drop fields inside the section (it will highlight).
- Repeat the same for each field.

### Section 1: Customer Information

- Customer Name
- Customer Type
- Email
- Phone
- Is Active

### Section 2: Address Details

- Address
- City
- State
- Pincode

9. Edit the layout as needed
10. Click Save.

# Activity 2: Edit Page Layout for Shipment__c

### Steps:

- Setup → Object Manager → Shipment__c
- Click Page Layouts
- Open the default Shipment Layout

### Sections to Add:

Section 1: Shipment Details

- Tracking Number
- Customer
- Shipment Type
- Weight
- Price
- Status

### Section 2: Address Information

- Source Address
- Destination Address

### Section 3: Delivery Information

- Expected Date
- Assigned Agent
- Arrange fields properly
- Click Save

# Activity 3: Edit Page Layout for Delivery_Agent__c

**Steps:**

- Setup → Object Manager → Delivery_Agent__c
- Page Layouts
- Open default Delivery Agent Layout

**Arrange Fields:**

**Agent Details**

- Agent Name
- Email
- Phone
- Branch
- Status
- Joining Date

**Additional Information**

- Related Shipments
- Related Delivery Updates
- Edit → Save

## Activity 4: Edit Page Layout for Branch__c

**Steps:**

- Setup → Object Manager → Branch__c
- Page Layouts
- Open default Branch Layout

**Sections:**

**Branch Information**

- Branch Name
- Branch Code
- City
- State
- Manager Name
- Arrange fields
- Click Save

## Activity 5: Edit Page Layout for Delivery_Update__c

**Steps:**

- Setup → Object Manager → Delivery_Update__c
- Page Layouts
- Open default Delivery Update Layout

**Sections:**

**Update Information**

- Update Number
- Shipment
- Status
- Update Date
- Remarks
- Arrange properly
- Click **Save**

## Activity 6: Edit Page Layout for Invoice__c

**Steps:**

- Setup → Object Manager → Invoice__c
- Page Layouts
- Open default Invoice Layout

**Sections:**

**Invoice Details**

- Invoice Number
- Shipment
- Amount
- Invoice Date

- Payment Status
- Arrange fields as needed
- Click Save

# Milestone 7 - Flows

In Salesforce, a flow is a powerful tool that allows you to automate business processes, collect and update data, and guide users through a series of screens or steps. Flows are built using a visual interface and can be created without any coding knowledge.

## Activity 1: Auto-Update Shipment Status When Delivery Update is Created

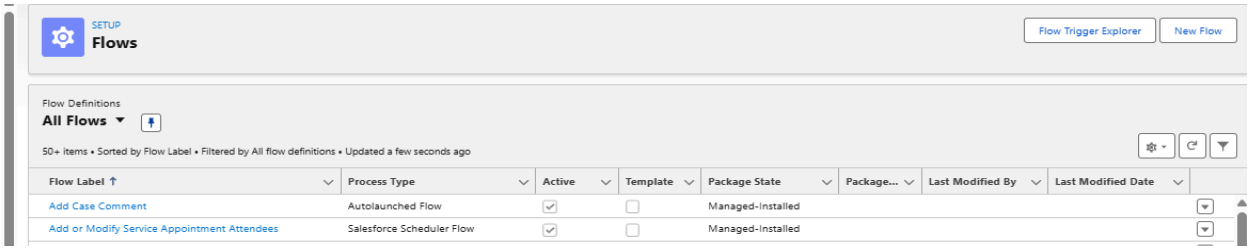
### Scenario:

Whenever a Delivery_Update__c record is created, automatically update the related Shipment__c Status with the latest status from Delivery Update.

### Part 1: Create the Record-Triggered Flow

### Step 1: Open Flow Builder

- Click the Gear Icon → Click Setup
- In the Quick Find box (left side), type Flows
- Click Flows
- Click the New Flow button




- Select Record-Triggered Flow
- Click Create

### Step 2: Configure the Start (Trigger)

On the Start panel on the right:

- Object: select Delivery_Update__c
- Trigger the Flow When:
    - Select A record is created
- Condition Requirements:
    - Choose: None (Always Run the Flow)
- Optimize the Flow For:
    - Select Actions and Related Records
- Click Done

## Part 2: Update Related Shipment Record

### Step 3: Add “Update Records” Element

- Click the + icon under the Start element
- Select Update Records

### Configure the Update Records Element:

- Label: Update_Shipment_Status
- How to Find Records to Update:
    - Select Specify conditions to identify records

### Set Filter Conditions:

- Object: Shipment__c
- Field: Id
- Operator: Equals
- Value: {!$Record.Shipment__c}

(This means update the Shipment linked to the Delivery Update)

### Set Field Values to Update

Add field mapping:

| Shipment Field | Value Source |
|---|---|
| Status__c | {!$Record.Status__c} |

(This copies status from Delivery Update to Shipment)

- Click Done

### Part 3: Save and Activate

### Step 4: Save Flow

- Click Save (top-right)
- Flow Label: Auto_Update_Shipment_Status
- API Name will auto-fill
- Click Save

### Step 5: Activate Flow

- Click Activate
- Flow status should show Active

## Milestone 8 - The Lightning App :

### What is an App ?

Apps in Salesforce are a group of tabs that help the application function by working together as a unit. It has a name, a logo, and a particular set of tabs. The simplest app usually has just two tabs.

# Activity 1: Create a Lightning App

### Create the Courier Management System App

1. From Setup, enter App Manager in the Quick Find and select App Manager.
2. Click New Lightning App.

    - Enter the Courier Management System as the App Name.
3. Upload an image or icon for the app, then click Next.
4. Under App Options, leave the default selections and click Next.
5. Under Utility Items, leave as it is and click Next.
6. From Available Items, select the following and move them to Selected Items:
    - Customer
    - Shipment
    - Delivery Agent
    - Branch
    - Delivery Update
    - Invoice
    - Reports
    - Dashboards
7. Click Next.
8. From Available Profiles, select System Administrator and move it to Selected Profiles.
9. Click Save & Finish.

# Milestone 9- Profile:

**What is a profile?**

A profile is a group/collection of settings and permissions that define what a user can do in salesforce. A profile controls “Object permissions, Field permissions, User permissions, Tab

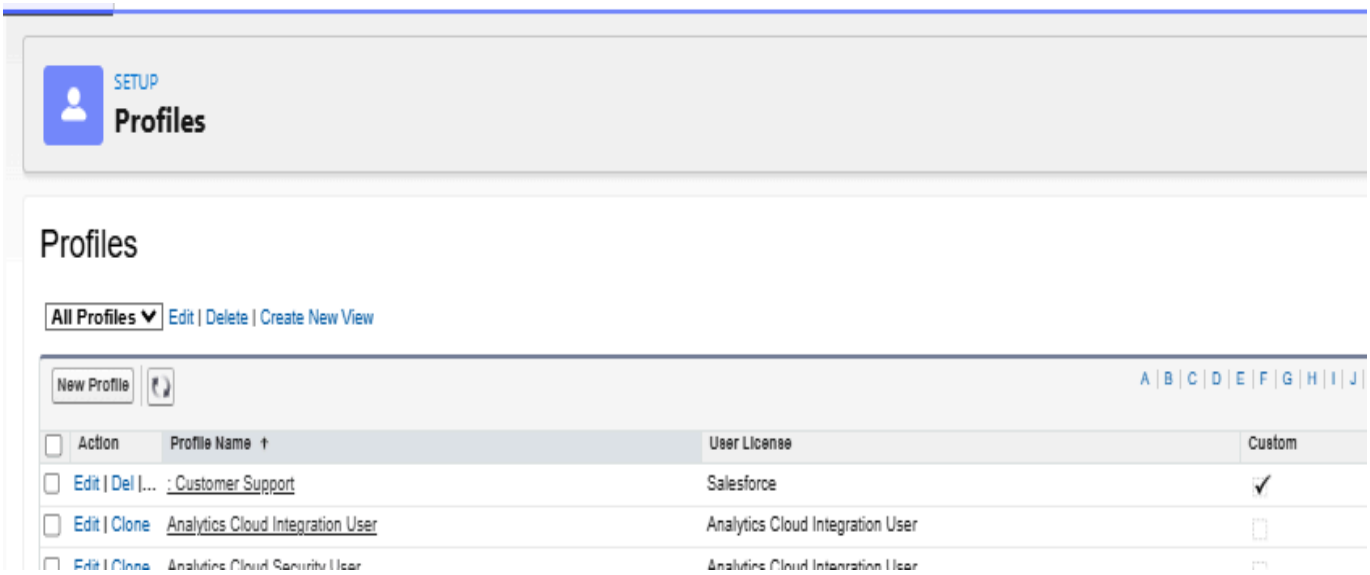
settings, App settings, Apex class access, Visualforce page access, Page layouts, Record Types, Login hours & Login IP ranges.

A profile can be assigned to many users, but users can be assigned a single profile at a time.

# Activity 1: Create a Custom Profile – Courier Admin

### Create a Custom Profile:

1. From Setup, enter Profiles in the Quick Find box.
2. Click Profiles.
3. Click New Profile.




4. Select Existing Profile as: System Administrator.
5. Enter Profile Name: Courier Admin.
6. Click Save.

### Why use System Administrator as Base?

It already has:

- Modify All Data
- Customize Application
- Full CRUD access on all objects

By cloning it, the **Courier Admin profile automatically inherits full access**.
You can later restrict permissions if required.

# Activity 2: Create a Custom Profile – Branch Manager

## Steps:

1. From Setup, enter Profiles in the Quick Find box.
2. Click Profiles.
3. Click New Profile.
4. Select Existing Profile as: Standard Platform User.
5. Enter Profile Name: Branch Manager.
6. Click Save.
7. Click Edit on the Branch Manager profile.

## Set App & Tab Settings

Under Assigned Apps / Custom App Settings, set the default app to:

Courier Management System App

## Ensure Visible Tabs Include:

Set these tabs as Default On or Visible:

- Customers
- Shipments

- Delivery Agents
- Branches
- Delivery Updates
- Invoices
- Reports
- Dashboards

## Custom Object Permissions

Scroll to Custom Object Permissions and set access as follows:

| Object | Permissions to Grant |
|---|---|
| Customer__c | Create, Read, Edit |
| Shipment__c | Create, Read, Edit |
| Delivery_Agent__c | Read, Edit |
| Branch__c | Read, Edit |
| Delivery_Update__c | Create, Read, Edit |
| Invoice__c | Read |

### Session & Password Settings

Under Session Settings:

- Session Timeout: 2 Hours of Inactivity

Under Password Policies:

- User passwords expire in: Never Expires
- Minimum Password Length: 8

Click Save.

# Activity 3: Create a Custom Profile – Delivery Agent

### Steps:

1. Setup → Quick Find → Profiles
2. Click New Profile
3. Existing Profile: Standard Platform User
4. Profile Name: Delivery Agent
5. Click Save

### App & Tab Access

Default App: Courier Management System

Visible Tabs:

- Shipments
- Delivery Updates

**Object Permissions :**

| Object | Permissions |
|---|---|
| Shipment__c | Read, Edit |
| Delivery_Update__c | Create, Read, Edit |
| Customer__c | Read |
| Invoice__c | Read |

### Restrictions

- No access to Branch editing
- No delete permissions
- Limited reporting access

Click Save.

# Activity 4: Create a Custom Profile – Customer Support

### Steps:

- Setup → Profiles → New Profile
- Base Profile: Standard User
- Profile Name: Customer Support

### Access Configuration

Tabs Visible:

- Customers
- Shipments
- Delivery Updates

## Object Permissions :

| Object | Access |
|---|---|
| Customer__c | Read, Edit |
| Shipment__c | Read, Edit |
| Delivery_Update__c | Read |
| Invoice__c | Read |

## Session Settings

- Session Timeout: 2 Hours
- Password Length: 8

Click Save

## Final Profiles Created in Project :

| Profile Name | Purpose |
|---|---|

| | |
|---|---|
| Courier Admin | Full system control |
| Branch Manager | Manage shipments and agents |
| Delivery Agent | Update delivery status |
| Customer Support | Handle customers and tracking |

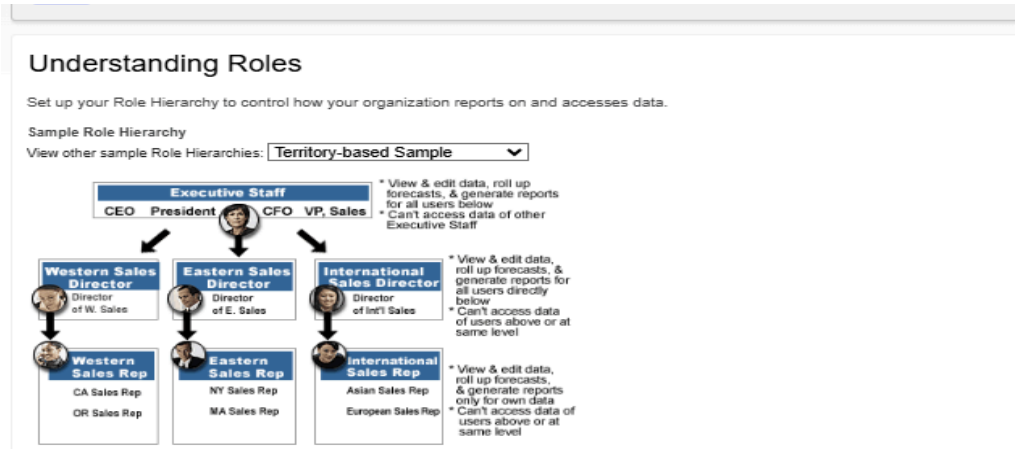
# Milestone 10- Role:

In Salesforce, roles are used to determine which users have access to certain data and functions within the system. They are also used to define the reporting hierarchy within an organization. Users with higher roles have greater access to data and more control over the system.

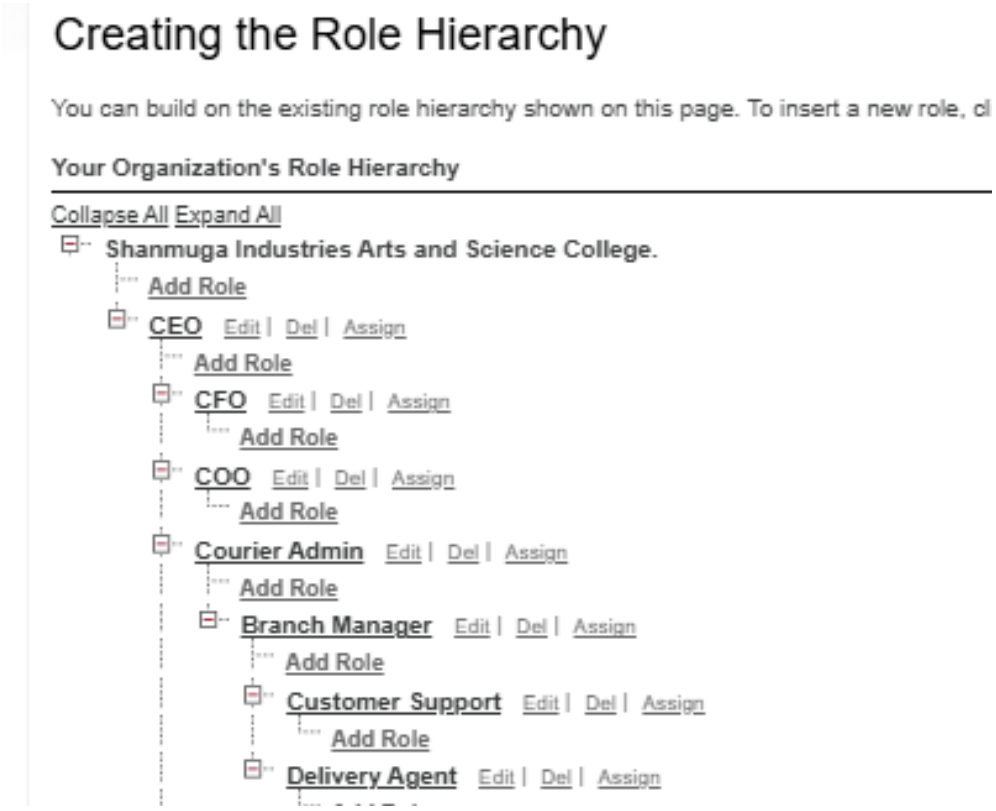
## Activity 1: Creation of Courier Admin Role

### Creation of Role:

1. From the Quick Find box, search for Roles and click on the Roles option.
2. Select Set Up Roles.



3. Click on Expand All and click Add Role under CEO.




4. Give Label as “Courier Admin”.
    - The Role Name will auto-populate.
5. Click Save.

# Activity 2: Creating Branch Manager Role Under Courier Admin Role

### Create another role:

1. Create a role with the name “Branch Manager”.
2. Below the Courier Admin, click on Add Role.
3. Enter the label name as “Branch Manager” and the Role Name will automatically populate.

4. Click Save.

# Activity 3: Creating Delivery Agent Role Under Branch Manager Role

### Steps:

1. Below the Branch Manager role, click on Add Role.
2. Enter Label as “Delivery Agent”.
    - Role Name will auto-populate.
3. Click Save.

# Activity 4: Creating Customer Support Role Under Branch Manager

### Steps:

1. Under the Branch Manager role, click on Add Role.
2. Enter Label as “Customer Support”.
3. Role Name will automatically populate.
4. Click Save.

### Purpose of This Hierarchy :

- **Courier Admin** – Highest level access to all data
- **Branch Manager** – Manages branch-level operations
- **Delivery Agent** – Access only to assigned shipments
- **Customer Support** – Limited access to customer and shipment tracking

This hierarchy ensures proper data sharing and security in the Courier Management System.

## Milestone 11- User :

**What is a user?**

A user is anyone who logs in to Salesforce. Users are employees at your company, such as sales,reps, managers, and IT specialists, who need access to the company's records. Every user in Salesforce has a user account. The user account identifies the user, and the user account settings determine what features and records the user can access.

**NOTE-** As Salesforce license can only be used by 2 Users at a time in Dev Org, So If you don't find a salesforce license then deactivate a user who has a salesforce license Or change the license type from Salesforce to any other.

# Activity 1: To Create a User

### Steps:

1. From Setup, enter Users in the Quick Find box, then select Users.
2. Click New User.

### Fill in the Fields:

| Field Name | Value |
|---|---|
| First Name | Enter first name (Example: Ravi) |
| Last Name | Enter last name (Example: Kumar) |
| Alias | Give an alias name |
| Email ID | Enter your personal email ID |
| Username | Must be in the format: text@text.text and must be unique |
| Nickname | Give a nickname |
| Role | Select **Courier Admin** |
| User License | Salesforce |

| | |
|---|---|
| Profile | **Courier Admin** |

3. Click Save.

## Activity 2: Creating Other Users

Repeat the same steps as **Activity 1** for creating additional users.

Use the following details:

| User Type | Role | User License | Profile |
|---|---|---|---|
| Branch Manager | Branch Manager | Salesforce | Branch Manager |
| Delivery Agent | Delivery Agent | Salesforce Platform | Delivery Agent |
| Customer Support | Customer Support | Salesforce Platform | Customer Support |

### Example:

To create a Branch Manager User:

- Go to Setup → Users → New User
- Fill details like:

| Field | Example Value |
|---|---|
| First Name | Arjun |
| Last Name | Reddy |
| Email | arjun@gmail.com |
| Username | arjun.manager@cms.com |
| Role | Branch Manager |

| | |
|---|---|
| Profile | Branch Manager |

- Click Save

\

### To Create Delivery Agent User:

Follow same steps with:

- Role: Delivery Agent
- Profile: Delivery Agent

### To Create Customer Support User:

- Role: Customer Support
- Profile: Customer Support

### Important Points

- Every user must have:
    - Unique username
    - Valid email
    - Assigned Role
    - Assigned Profile
- Profiles control permissions
- Roles control data visibility

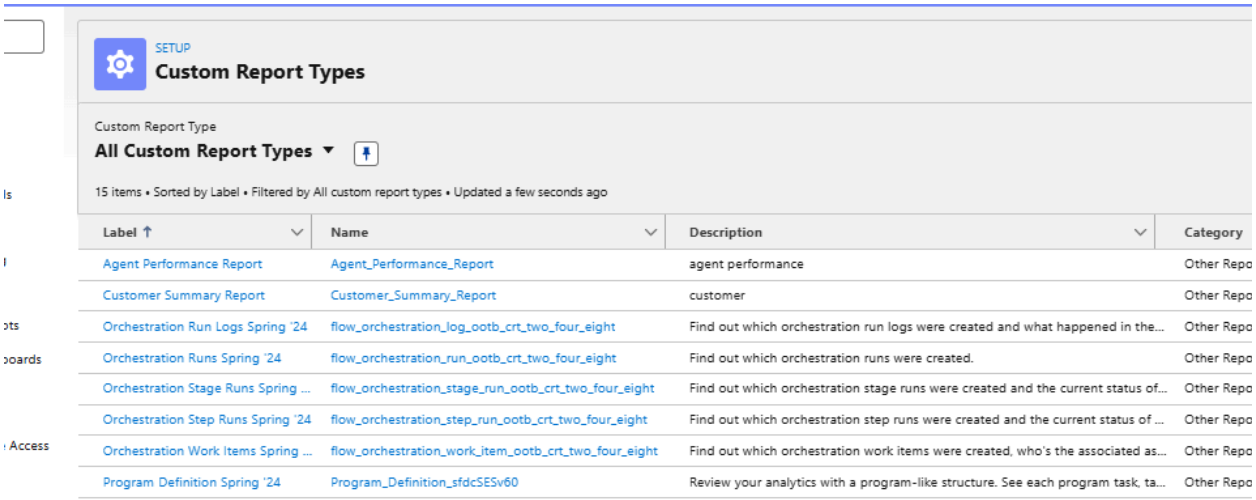
# Milestone 15 : Reports & Dashboards

Reports and Dashboards help analyze business data and track courier operations efficiently.

## Activity 1: Create Report Types

### Step 1: Open Report Types

1. Click Gear Icon → Select Setup.
2. In Quick Find search Report Types.
3. Click New Custom Report Type.




### Step 2: Create Shipment Report Type

- Primary Object: Shipment__c
- Label: Shipment Tracking Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

### Step 3: Create Customer Report Type

- Primary Object: Customer__c
- Label: Customer Summary Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

### Step 4: Create Agent Performance Report Type

- Primary Object: Delivery_Agent__c
- Label: Agent Performance Report
- Category: Reports
- Deployment Status: Deployed
- Click Next → Save

# Activity 2: Create Reports

### Report 1: Shipment Status Summary Report

### Step 1: Create the Report

1. Click App Launcher (9 dots) → Search Reports
2. Click New Report
3. Choose Shipment Tracking Report
4. Click Continue

### Step 2: Add Columns

Add these fields:

- Tracking Number
- Customer
- Status
- Expected Date
- Assigned Agent

**Step 3: Add Filters**

- Show Me: All Shipments
- Date Filter: This Month
- Status Filter: Select specific Status (Optional)

**Step 4: Group & Summarize**

- Group rows by Status__c
- Add summary on Price

**Step 5: Save Report**

- Click Save & Run
- Name: Shipment Status Summary
- Folder: CMS Reports

**Report 2: Agent Performance Report**

**Step 1: Create Report**

- New Report → Choose Agent Performance Report
- Click Continue

### Step 2: Add Columns

- Agent Name
- Branch
- Number of Shipments
- Status

### Step 3: Group Data

- Group by Delivery_Agent__c

### Step 4: Save Report

- Name: Agent Performance Report
- Folder: CMS Reports
- Save & Run

### Report 3: Revenue Report

### Steps

- Create report using Shipment Tracking Report Type

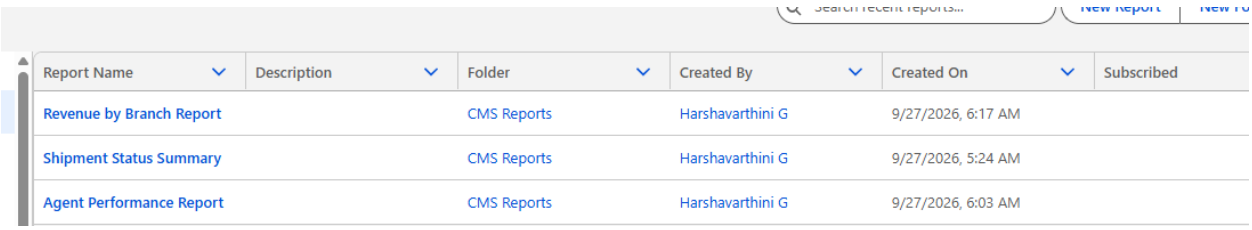
Add fields:

- Tracking Number
- Customer
- Price
- Invoice Status

Group by:

- Branch
- Month

Save as: Revenue by Branch Report



| Report Name | Description | Folder | Created By | Created On | Subscribed |
|---|---|---|---|---|---|
| Revenue by Branch Report | | CMS Reports | Harshavarthini G | 9/27/2026, 6:17 AM | |
| Shipment Status Summary | | CMS Reports | Harshavarthini G | 9/27/2026, 5:24 AM | |
| Agent Performance Report | | CMS Reports | Harshavarthini G | 9/27/2026, 6:03 AM | |

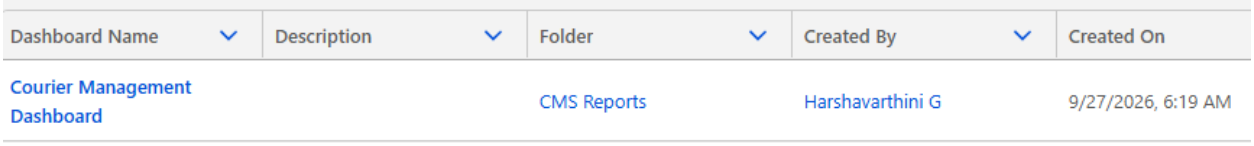
# Activity 3: Create Dashboard

## Step 1: Open Dashboards

1. Click App Launcher (9 dots)
2. Search Dashboards
3. Click New Dashboard

## Step 2: Enter Dashboard Details

- Name: Courier Management Dashboard
- Folder: CMS Reports
- Click Create



| Dashboard Name | Description | Folder | Created By | Created On |
|---|---|---|---|---|
| Courier Management Dashboard | | CMS Reports | Harshavarthini G | 9/27/2026, 6:19 AM |

# Activity 4: Add Dashboard Components

## Component 1: Shipment Status Overview (Donut Chart)

1. Click + Component
2. Choose report: Shipment Status Summary
3. Visual Type: Donut Chart
4. Click Add

## Component 2: Agent Performance (Bar Chart)

- Select report: Agent Performance Report
- Chart Type: Bar Chart
- Click Add

## Component 3: Revenue by Branch (Column Chart)

- Choose report: Revenue by Branch Report
- Chart Type: Column Chart
- Click Add

# Activity 5: Save & Activate Dashboard

1. Click Save
2. Click Done

**Share Dashboard With:**

- Courier Admin
- Branch Managers
- Customer Support Team

## 7. FUNCTIONAL & PERFORMANCE TESTING

During testing, screenshots were captured from Salesforce to verify record creation, validation rule enforcement, delivery status updates, flow automation, invoice generation, and dashboard visualization. All scenarios were validated to ensure correct performance and reliability.

## 8. ADVANTAGES & DISADVANTAGES

**Advantages**

- Centralized data
- Automated workflows
- Real-time reporting

**Disadvantages**

- Salesforce dependency
- Requires initial configuration effort

## 9. Conclusion:

The Courier Management System successfully digitizes courier operations using Salesforce automation, improving efficiency, accuracy, and customer satisfaction.

# 10. FUTURE SCOPE

- Mobile tracking app
- GPS integration
- SMS notifications
- AI-based delivery optimization

# 11. APPENDIX

- Source Code: Salesforce Configuration
- Dataset: Salesforce Objects
- GitHub / Demo: Salesforce Developer Org

## Thank You